REST API Specification & Reference Manual

Healthcare Patient Management Microservice | API Version 1.0.0

Authentication: HTTP Bearer JWT | Host: 127.0.0.1:8000 | OpenAPI Docs: /docs

POST /api/v1/auth/login	Method: POST
Description: Authenticate user credentials using Bcrypt and issue signed JWT bearer token.	
Sample Request: {"username": "clinician", "password": "ClinicianPass123!"}	**Sample Response:** {"access_token": "eyJhbG...", "token_type": "bearer", "role": "clinical_operations"}

GET /api/v1/patients	Method: GET
Description: Retrieve paginated list of patients with search and demographic filters.	
Sample Request: Params: page=1&page;_size=50&gender;=Female■■=Caucasian	**Sample Response:** {"total": 71520, "page": 1, "items": [{"patient_nbr": 8222157, "race": "Caucasian", "gender": "Female"}]}

POST /api/v1/patients	Method: POST
Description: Create a new patient record (Requires Clinical Operations or Administrator role).	
Sample Request: {"patient_nbr": 88888001, "race": "Hispanic", "gender": "Female", "age_bracket": "[50-60)"}	**Sample Response:** {"patient_nbr": 88888001, "created_at": "2026-10-11T12:00:00"}

GET /api/v1/encounters	Method: GET
Description: Search encounters with sorting, pagination, LOS ranges, and readmission status.	
Sample Request: Params: readmitted=<30&min;_los=3&sort;_by=admission_date&sort;_order=desc	**Sample Response:** {"total": 11357, "items": [{"encounter_id": 149190, "patient_nbr": 55629189, "time_in_hospital": 3}]}

PUT /api/v1/encounters/{id}	Method: PUT
Description: Update encounter details (time_in_hospital, specialty, medications, readmission). Audited.	
Sample Request: {"time_in_hospital": 4, "medical_specialty": "Cardiology", "num_medications": 15, "readmitted_raw": "<30"}	**Sample Response:** {"encounter_id": 149190, "status": "updated", "time_in_hospital": 4}

GET /api/v1/analytics/summary-kpis	Method: GET
Description: Retrieve real-time hospital KPIs (total encounters, 30d readmission rate, avg LOS).	
Sample Request: None (JWT optional in local mode)	**Sample Response:** {"total_encounters": 101766, "total_patients": 71520, "avg_length_of_stay": 4.40, "readmission_rate_30d_pct": 11.16}

GET /api/v1/analytics/readmission-by-category	Method: GET
Description: Calculate readmission percentages grouped by primary ICD-9 diagnosis category.	
Sample Request: None	**Sample Response:** {"by_diagnosis": [{"category": "Circulatory", "total": 30437, "readmission_rate_pct": 11.38}]}

POST /api/v1/etl/incremental	Method: POST
Description: Upload and incrementally ingest CSV batch with automated data quality checks.	

Sample Request: multipart/form-data; file=@batch.csv	**Sample Response:** {"batch_id": "batch_20261011_01", "records_ingested": 100, "status": "COMPLETED"}

GET /api/v1/reports/export/csv	**Method: GET**
Description: Download filtered encounters dataset as a formatted CSV file.	
Sample Request: Params: limit=5000	**Sample Response:** Binary text/csv stream attachment